

RCHAIN/RLOG Boundary: Custody and Event-Log Evidence

Gio Della-Libera

Draft — June 28, 2026

Abstract

Election count reconciliation can show that declared arithmetic is internally consistent, but it cannot prove the physical history of ballots, devices, media, seals, or rooms. This paper reserves two downstream civic-evidence layers for those process records without implementing either one. RLOG is the future owner of normalized event and status logs: what source logs record. RCHAIN is the future owner of chain-of-custody evidence: what custody sources record. Both layers are deliberately deferred until public source artifacts such as EMS logs, tabulator logs, seal logs, custody forms, and observer records force concrete schemas. The boundary rule is simple: RLOG can say what a source log records; RCHAIN can say what a custody source records; neither can infer custody or correctness from count reconciliation alone.

1 Introduction

Custody and event-log evidence matter because public trust is not only a count arithmetic question. Voters and observers also ask when equipment was used, which files were exported, whether media moved through expected hands, whether seals and containers matched the records, and whether an observed process fits the official explanation. Those are real election-administration questions, but they are physical-process claims.

RCOUNT is not the right owner for those claims. A count-reconciliation verifier can check that declared totals, subtotals, manifests, cast vote records, audit records, and package hashes reconcile under named rules. It cannot conclude from that reconciliation that ballots were always in proper custody, that a seal was never broken, that a tabulator log is complete, or that an operator performed an action for a stated reason.

This paper therefore reserves two downstream layers. RLOG is reserved for normalized event and status logs. RCHAIN is reserved for physical chain-of-custody evidence. The reservation is not an implementation claim. It is a boundary claim: RCOUNT should remain honest about count evidence, and the future custody and log layers should wait for source artifacts before they harden schemas.

2 Ownership Contract

The ownership contract separates two kinds of process evidence. RLOG is about records emitted by election systems or administrative processes. It can normalize what a source log records and bind that normalization to source bytes, parser transcripts, caveats, and optional RCOUNT identifiers. RCHAIN is about physical custody records. It can normalize what a custody source records: seals, transfers, observers, storage locations, retrieval sheets, and gaps.

Neither layer is allowed to convert count reconciliation into process proof. The distinction is small but important. A reconciled count package can be an input to a larger public evidence bundle, but it is not a custody record and it is not an event log.

Layer	May assert	Must not assert
RLOG	A source log records an event, status, timestamp, device identifier, operator label, export, upload, adjudication step, parser caveat, or link to a count artifact.	Physical custody, ballot security, legal compliance, or count correctness unless those facts are explicitly present in the source log and stated only as source-record claims.
RCHAIN	A custody source records a seal, transfer, observer note, storage location, retrieval sheet, custody gap, or caveat.	Count correctness, certification, or an unrecorded custody fact inferred from successful RCOUNT reconciliation, statistics, or absence of a discrepancy.
RCOUNT	Count, manifest, CVR, audit, and reconciliation facts that follow from public count evidence.	Custody continuity, physical handling, device-event completeness, or administrative intent.

Table 1: Reserved ownership boundaries for process evidence.

3 Deferral and Boundary

The decision for X.3 is to defer both RLOG and RCHAIN. No RLOG crate, RCHAIN crate, production type, parser, fixture ladder, or verifier command is claimed here. The downstream-boundary and civic-evidence access-pattern specifications both place these packages after the base evidence layers and state that RLOG and RCHAIN should be added only when real log and custody source artifacts demand them [Della-Libera, 2026a].

That order matters. Public EMS logs, scanner logs, tabulator logs, upload logs, adjudication logs, seal logs, custody forms, ballot retrieval sheets, and observer records vary by jurisdiction and vendor. A schema written before those artifacts are collected would mostly encode guesses. X.3 therefore requires L0 source-pressure fixtures before any crate work: small positive and negative examples based on real public record shapes, with source bytes or source-like records, expected normalizations, and explicit caveats.

The boundary rule is the central contribution: RLOG can say what a source log records. RCHAIN can say what a custody source records. Neither can infer custody or correctness from count reconciliation alone. If an RCOUNT package balances perfectly, that is evidence

about the count contract. It is not by itself evidence that a seal was intact, that a room was continuously observed, that every machine event was exported, or that no custody gap occurred.

The same rule also limits future composition. A later RCASE bundle may cite RCOUNT, RLOG, and RCHAIN together, and RCERT may reference them when an official action depends on process evidence. But composition must preserve the child claim boundaries. It may say which package supports which public claim; it must not let a count verifier silently become a custody verifier.

4 Conclusion

X.3 reserves RLOG and RCHAIN so that custody and event-log evidence have clear owners when the project is ready for them. RLOG is reserved for normalized event and status logs. RCHAIN is reserved for physical chain-of-custody records. RCOUNT remains the owner of count reconciliation and must not claim physical process facts that its evidence cannot prove.

Deferral is the right call now. The missing work is not code polish; it is source pressure. The project still needs public examples of EMS logs, tabulator logs, seal logs, custody forms, and observer records, followed by L0 fixtures that force minimal schemas. Until that evidence exists, not implementing these layers is the honest technical decision.

References

- Gio Della-Libera. Civic evidence layer boundaries and access patterns. Technical report, Apportionment Research, 2026a. Architecture specifications: downstream evidence boundaries and civic evidence layer access patterns.
- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026b.